Microsoft Elevate

CAPSTONE PROJECT

GLOBAL SUPERSTORE SALES ANALYSIS

PRESENTED BY

STUDENT NAME: SACHIN PASWAN

COLLEGE NAME: IIT PATNA

DEPARTMENT: COMPUTER SCIENCE & DATA ANALYTICS

EMAIL ID: SACHINPASWAN2004@GMAIL.COM

OUTLINE:

- **Problem Statement** (Should not include solution)
- **Proposed System/Solution**
- **System Development Approach** (Technology Used)
- **Algorithm & Deployment**
- **Result (Output Image)**
- **Conclusion**
- **Future Scope**
- **References**

PROBLEM STATEMENT:

In modern retail businesses, a large volume of transactional data is generated from daily sales operations across different regions, customers, and product categories. However, this raw data is often complex and difficult for decision-makers to interpret without proper analysis and visualization. Organizations need an effective way to transform this data into meaningful insights that can support strategic decision-making.

The Global Superstore dataset contains detailed information about orders, sales, profit, discounts, shipping modes, and customer segments across multiple markets and countries. Without proper analysis, it becomes challenging to identify sales trends, profitable products, high-performing regions, and the impact of discounts and shipping costs on overall profitability.

Therefore, the main problem addressed in this project is how to analyze and visualize large retail sales data in an efficient and interactive way. By using Microsoft Power BI, this project aims to convert raw transactional data into a comprehensive dashboard that highlights key performance indicators, sales patterns, customer behavior, and profitability across different categories and regions. The goal is to help stakeholders easily understand business performance and support data-driven decision making.

PROPOSED SOLUTION:

- To address the challenge of analyzing complex retail sales data, a Business Intelligence solution was developed using **Microsoft Power BI**. The proposed solution focuses on collecting reliable data, processing it to ensure accuracy, and deploying an interactive dashboard that helps users easily understand business performance.

• 1. Data Collection

- The dataset used in this project was obtained from **Kaggle**, a widely used platform for open datasets and data science projects. The **Global Superstore Dataset** contains detailed information about orders, customers, products, sales, profit, discount, shipping modes, and geographical regions. This dataset provides sufficient information to analyze sales trends, profitability, and customer behavior across multiple markets and categories.

• 2. Data Processing

- After collecting the dataset, the data was cleaned and prepared using the Power Query Editor in **Microsoft Power BI**. Several data preprocessing steps were performed to ensure data quality and consistency. Duplicate records were removed to avoid incorrect analysis. Data types of different columns were corrected to ensure accurate calculations and analysis. Additionally, new calculated columns were created to support further analysis and improve the effectiveness of the dashboard. These steps helped transform the raw dataset into a structured and analysis-ready format.

• 3. Deployment

- Once the data preparation was completed, an interactive dashboard was created in **Microsoft Power BI**. The dashboard includes multiple visualizations such as KPI cards, bar charts, pie charts, maps, and slicers to analyze sales, profit, customers, products, and regional performance. Interactive filters such as Year, Region, and Segment were added to allow users to dynamically explore the data. The final dashboard provides clear insights into sales performance, profitability, and operational trends, enabling stakeholders to make informed and data-driven business decisions.

SYSTEM APPROACH:

The system approach of this project involves analyzing retail sales data and presenting insights through an interactive dashboard using Microsoft Power BI. The dataset was collected from Kaggle and then cleaned and prepared using Power Query, where duplicate records were removed, data types were corrected, and new columns were created for better analysis.

After data preparation, the processed dataset was used to create visualizations and key performance indicators such as sales, profit, and customer insights. Finally, an interactive dashboard was developed to help users easily understand business performance and support data-driven decision making.

ALGORITHM & DEPLOYMENT:

- Algorithm

- The algorithm used in this project follows a structured process for analyzing and visualizing retail sales data using Microsoft Power BI.
- Collect the Global Superstore dataset from Kaggle.
- Import the dataset into Power BI.
- Clean the data using Power Query by removing duplicate records and correcting data types.
- Create new columns and necessary calculations for analysis.
- Build data visualizations such as charts, maps, and KPI indicators.
- Apply filters and slicers to enable interactive data exploration.
- Analyze the results and generate insights from the dashboard.
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- Deployment

- The final dashboard was developed and deployed using Microsoft Power BI. After completing data cleaning, modeling, and visualization, the dashboard was organized into multiple report pages to present different aspects of the analysis such as sales performance, customer insights, product profitability, and regional trends. Interactive filters and visual elements were added to allow users to easily explore the data. The deployed dashboard provides a clear and user-friendly interface that helps stakeholders monitor key business metrics and make informed decisions based on data insights.

RESULT:

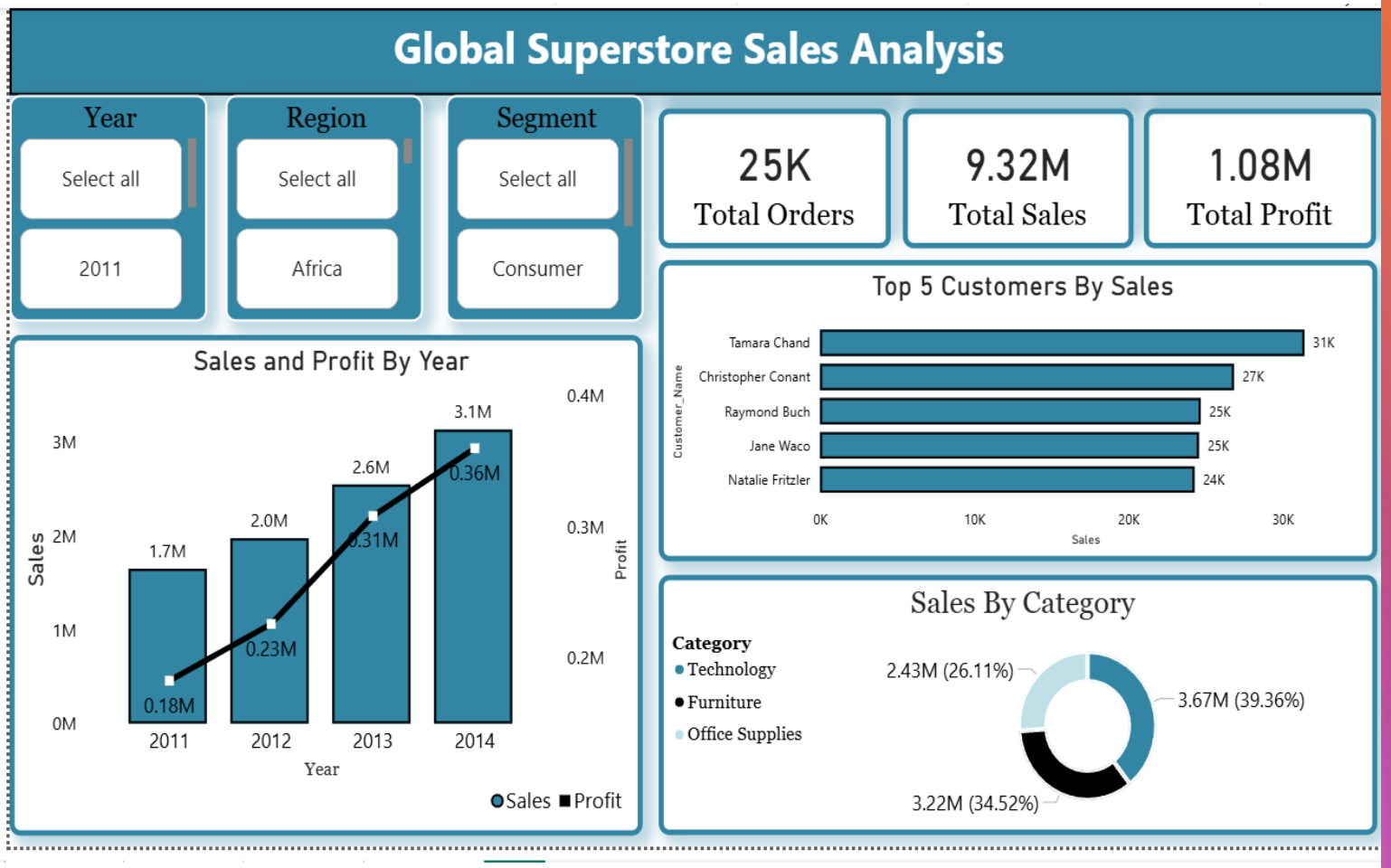
•Page 1: **Sales Overview Dashboard**

•**<u>Total Orders, Total Sales, Total Profit (KPI Cards)</u>**
These KPI cards provide a quick summary of overall business performance by showing the total number of orders, total sales revenue, and total profit.

•**<u>Sales and Profit by Year</u>**
This chart shows the yearly trend of sales and profit, indicating a steady growth in both metrics from 2011 to 2014.

•**<u>Top 5 Customers by Sales</u>**
This bar chart identifies the top five customers who generated the highest sales for the company.

•**<u>Sales by Category</u>**
This donut chart displays the distribution of sales across different product categories such as Technology, Furniture, and Office Supplies.

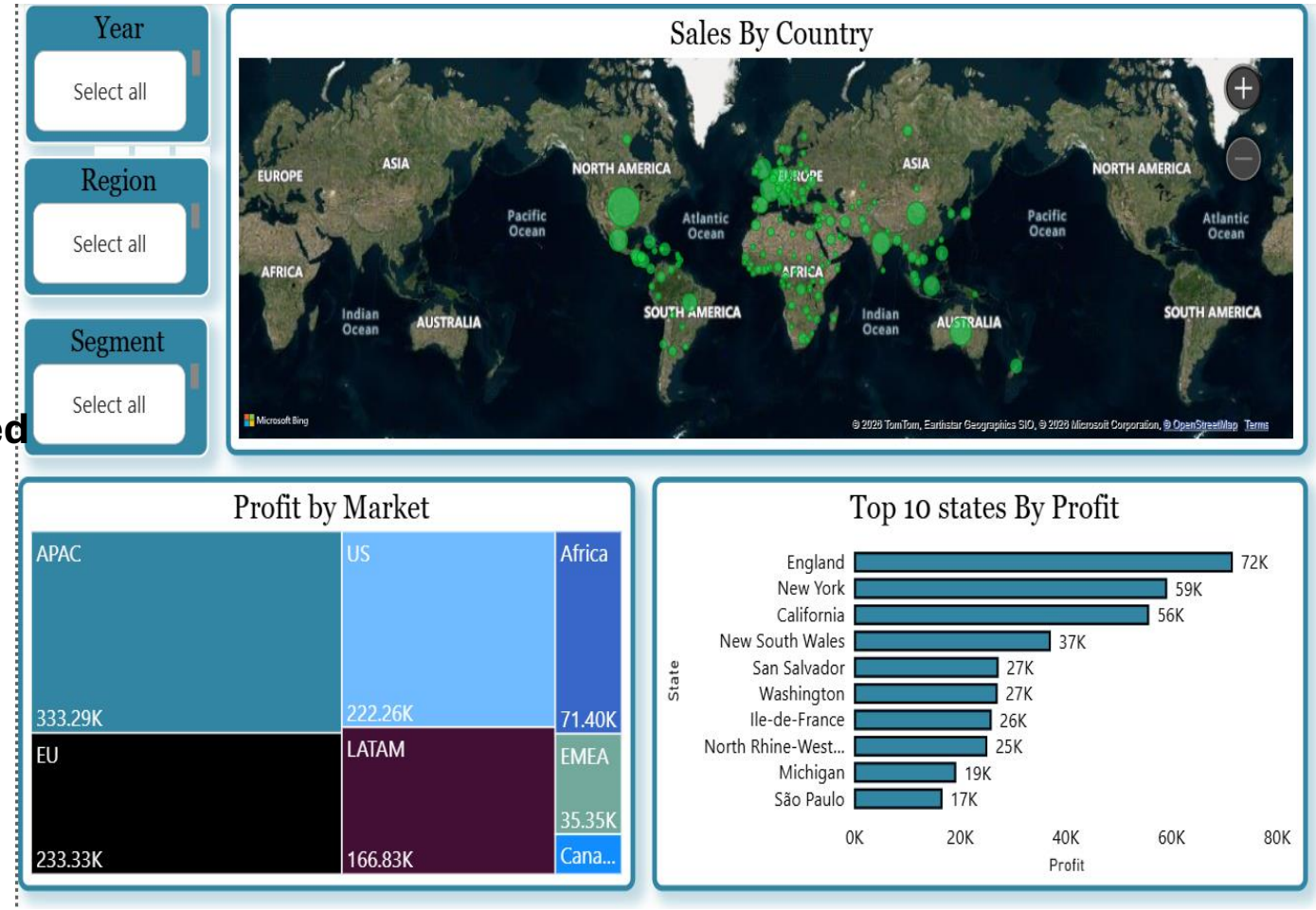


Regional and Market Analysis

- **Sales by Country (Map)**
  This map visualization shows the geographical distribution of sales across different countries around the world.

- **Profit by Market**
  This treemap illustrates how profit is distributed across different global markets such as APAC, EU, US, and LATAM.

- **Top 10 States by Profit**
  This bar chart highlights the top ten states that generate the highest profit for the business.



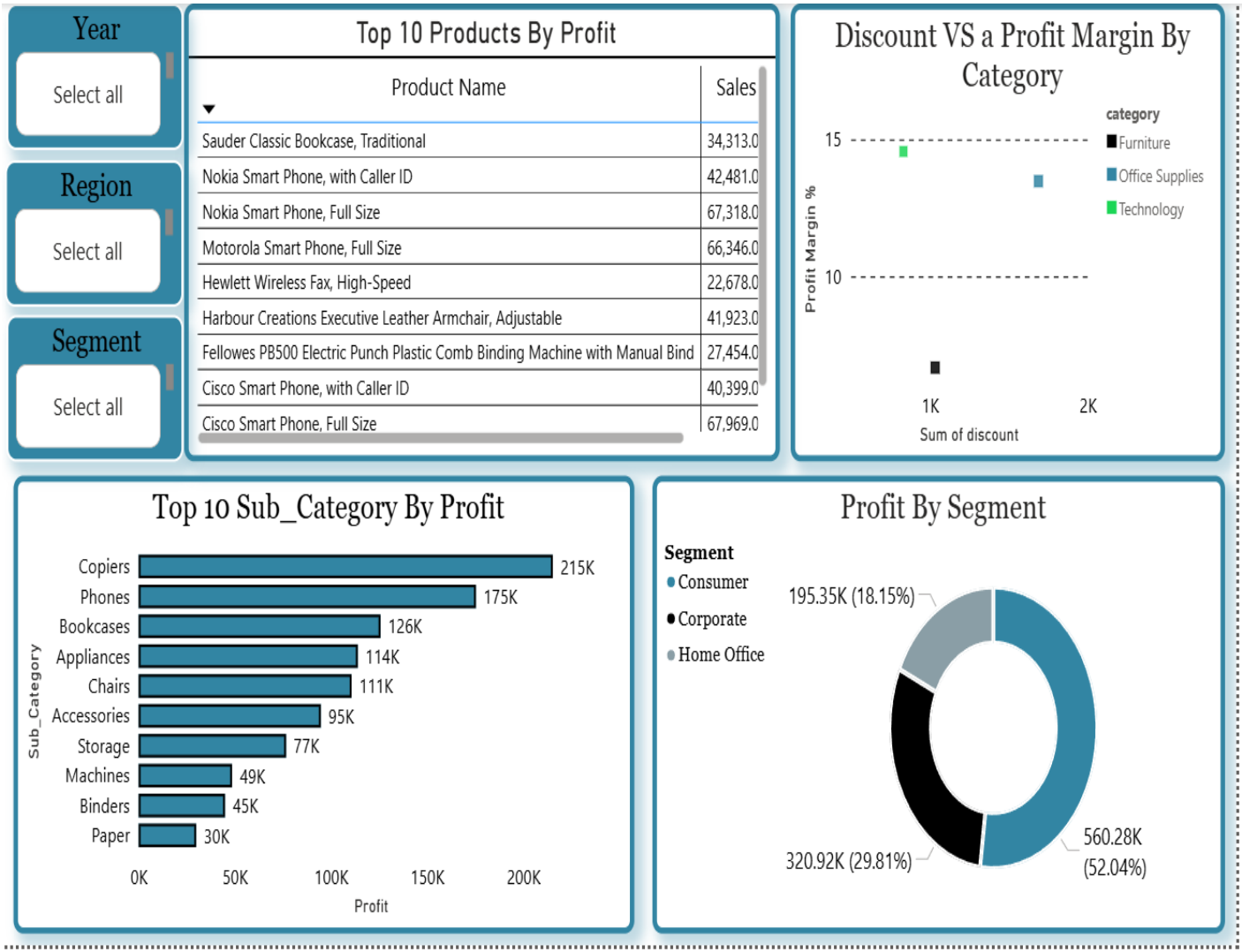
Product and Customer Insights

- **Top 10 Products by Profit**
  This table visualization lists the top ten products that contribute the most to the company's overall profit.

- **Discount vs Profit Margin by Category**
  This scatter chart analyzes the relationship between discount levels and profit margins across different product categories.

- **Top 10 Sub-Category by Profit**
  This bar chart shows which product sub-categories generate the highest profit.

- **Profit by Segment**
  This donut chart illustrates how profit is distributed among different customer segments such as Consumer, Corporate, and Home Office.



Shipping and Operational Analysis

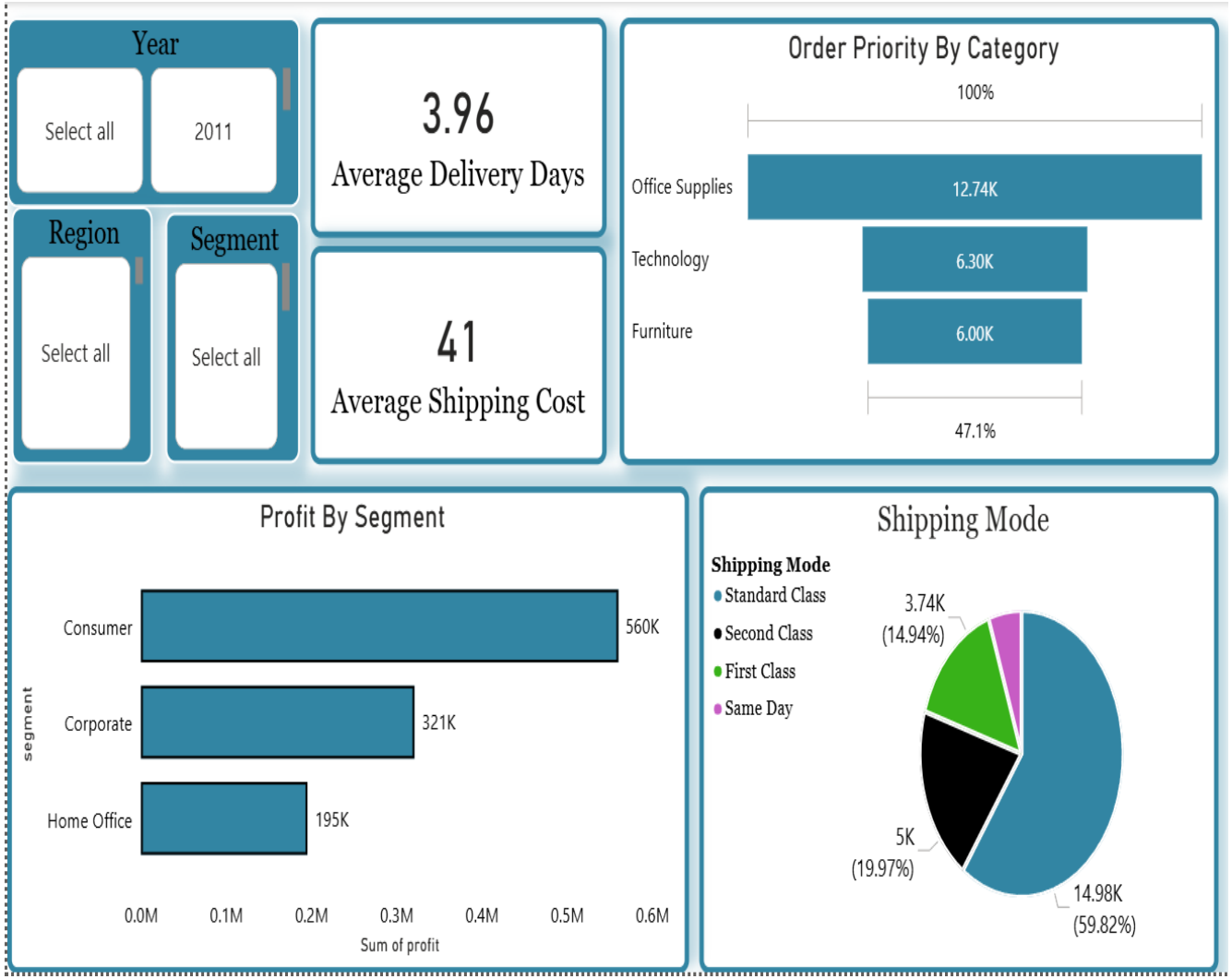
- **Average Delivery Days**
  This KPI shows the average number of days required to deliver orders to customers.

- **Average Shipping Cost**
  This KPI represents the average shipping cost incurred for delivering orders.

- **Order Priority by Category**
  This bar chart shows how order priorities are distributed across different product categories.

- **Profit by Segment**
  This chart compares the profit contribution of different customer segments.

- **Shipping Mode**
  This pie chart illustrates the proportion of orders delivered through different shipping modes such as Standard Class, Second Class, First Class, and Same Day.



CONCLUSION:

- This project successfully analyzed retail sales data from the **Global Superstore Dataset** and transformed raw data into meaningful business insights using **Microsoft Power BI**. By applying data cleaning, transformation, and visualization techniques, the project developed an interactive dashboard that provides a clear overview of sales performance, profitability, customer behavior, and regional trends.
- The analysis helped identify key factors influencing business performance, such as top-performing products, profitable customer segments, and high-revenue regions. Additionally, the dashboard highlights the impact of discounts, shipping modes, and delivery efficiency on overall profitability.
- Overall, the project demonstrates how business intelligence tools can convert complex datasets into easy-to-understand visual insights, enabling organizations to make more informed, data-driven decisions and improve their operational strategies.

FUTURE SCOPE:

In the future, this project can be improved by integrating real-time data sources and advanced analytics features. Predictive analysis and sales forecasting can also be added to better understand future business trends. Additionally, the dashboard created using **Microsoft Power BI** can be published on **Power BI Service** to enable online access, sharing, and collaboration among users.

REFERENCES:

DATA SET LINK

GITHUB LINK

Thank You